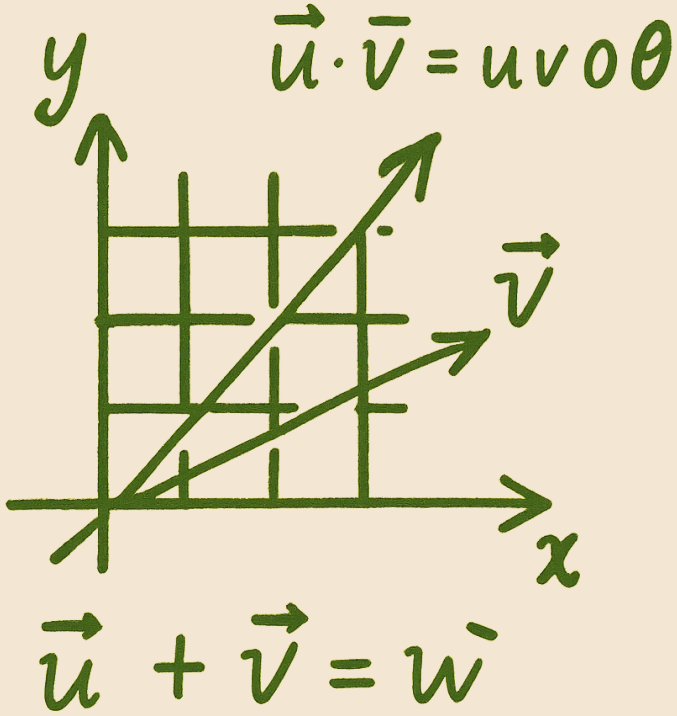


# Linear Algebra: The Foundation of Modern Computing

A B.Tech introduction to the mathematical language powering AI, data science, and computer graphics.

B.TECH · MATHEMATICS · SEMESTER 1

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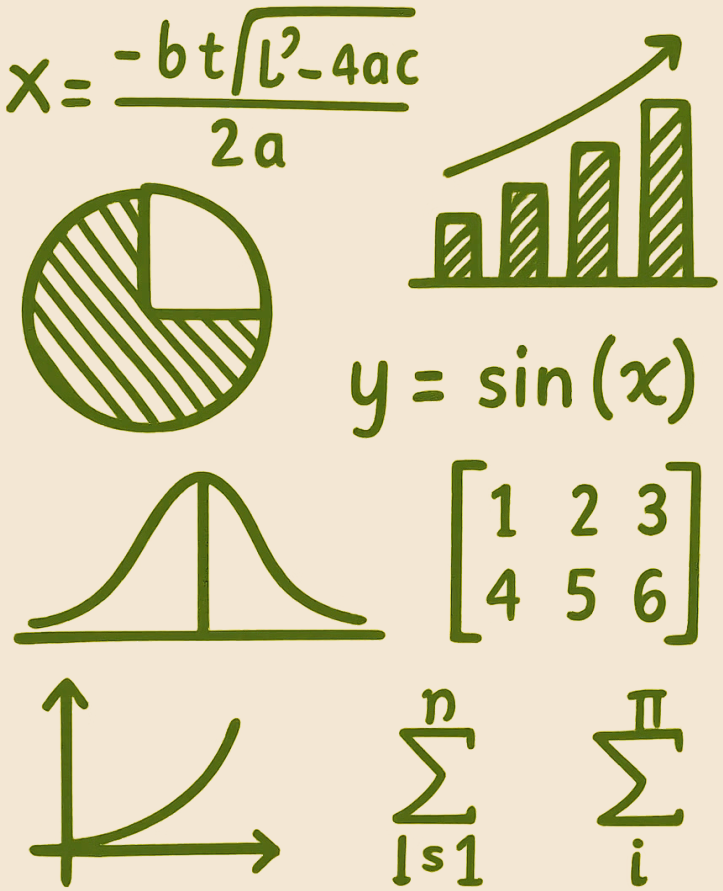


# What is Linear Algebra?

## More Than Numbers

Linear algebra is the **language of data and transformations** — a framework for describing how quantities relate, scale, and move through space.

## Why It Matters

It is the backbone of **Data Science, Artificial Intelligence, Machine Learning,** and **Computer Graphics**. Every model you train or render you see relies on these principles.

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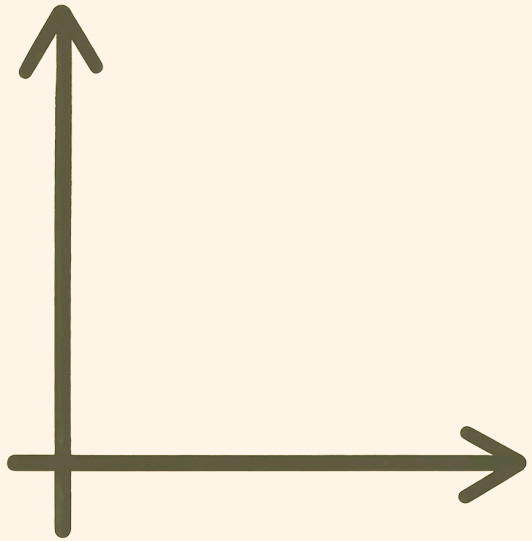

CORE CONCEPT

# Vectors: The Building Blocks

In Computer Science, a **vector** is an ordered list of numbers representing a point in n-dimensional space — visualised as an arrow from the origin.

### 2D Example

$v = [x_1,\ y_1]$ — a point in the 2D plane.

### n-Dimensional

$v = [x_1,\ x_2,\ \ldots,\ x_n]$ — generalises to any number of features.

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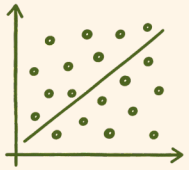
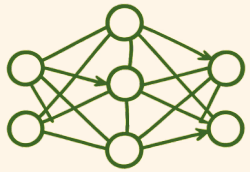
# Vectors in Data Science

Every dataset you work with is, at its core, a collection of vectors.



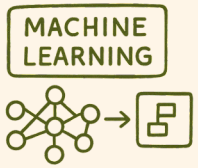
## Datasets as Points

Each data sample is an n-dimensional vector — every feature becomes one dimension in the space.



## Models as Vectors

Neural network weights, gradients, and parameters are all stored and manipulated as vectors.




## Training Examples

Each training example fed into a model is a vector — similarity and distance are computed using vector operations.

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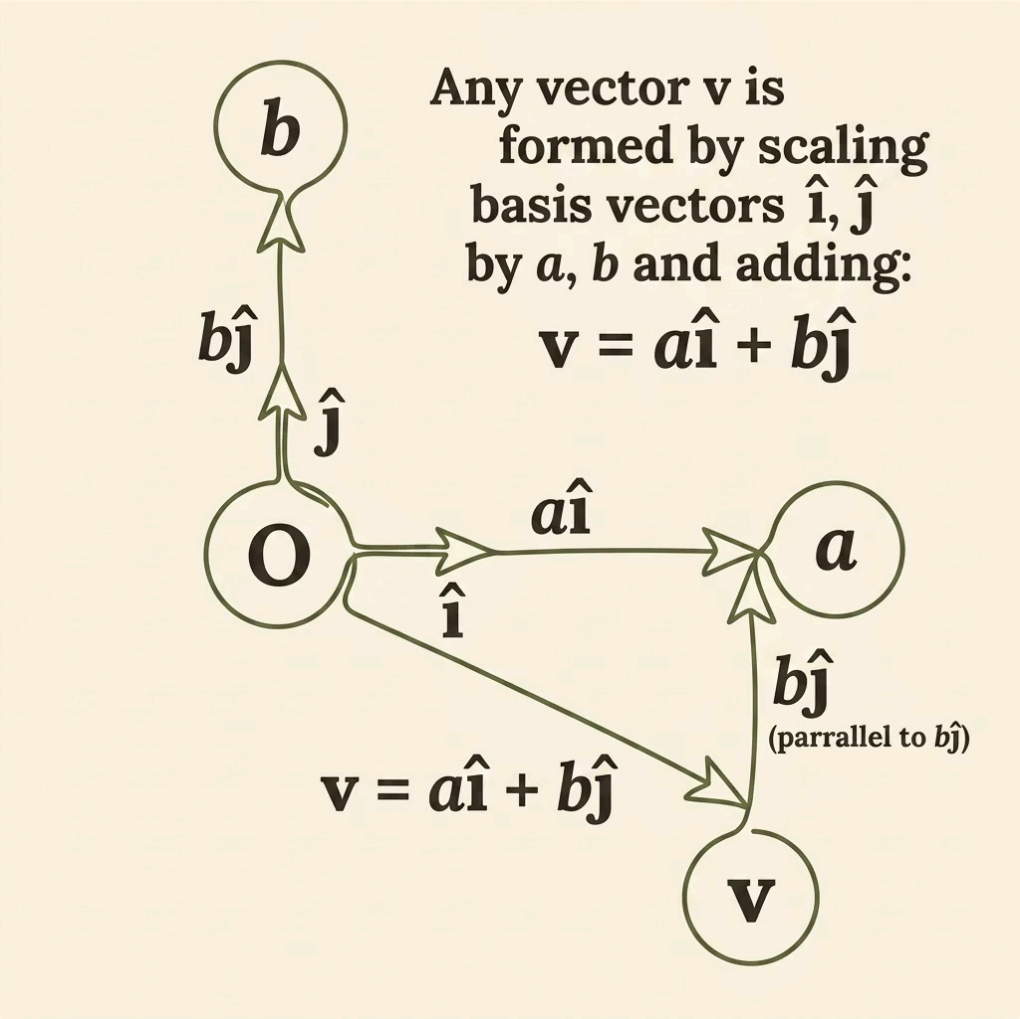
# Basis Vectors & Linear Combinations

**Basis vectors** (like $\hat{x}$ and $\hat{y}$) are the fundamental unit directions along each axis. **Any vector** in the space can be built by scaling and adding these basis vectors.

> **Key Formula:** $\vec{v} = x_1\hat{x} + y_1\hat{y}$ — every vector is a **linear combination** of its basis.

This concept generalises to any number of dimensions and is the foundation of coordinate systems.

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# Visualising Data with Vectors

## Data Points as Vectors

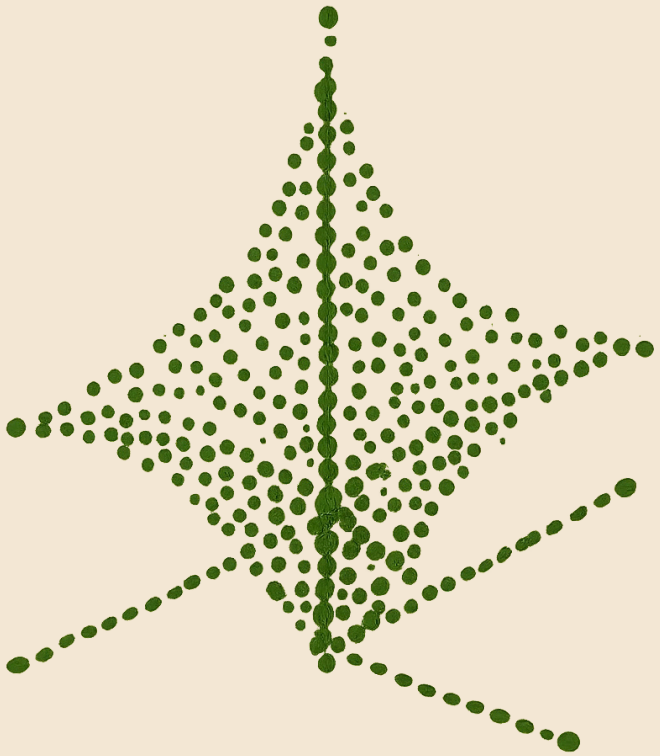
Each observation in your dataset is a single vector — a point in high-dimensional feature space.

## The Dataset as a Cloud

The full dataset forms a **point cloud** — patterns, clusters, and separations become visible geometrically.

## Flexible Basis

Basis vectors need not align with axes — choosing a different basis can simplify or reveal structure in data.



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# Groups, Rings & Fields

These **algebraic structures** define the rules governing mathematical operations — the formal groundwork of linear algebra.

### Group

A set with one binary operation satisfying **closure, associativity, identity,** and **inverse.**

### Ring

Extends a group with a **second operation** (multiplication) that distributes over addition.

### Field

A ring where every non-zero element has a **multiplicative inverse.** Example: Real Numbers $\mathbb{R}$.

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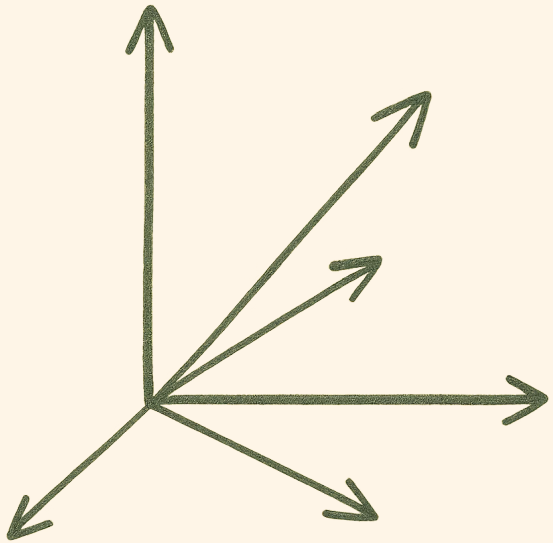

CORE STRUCTURE

# Vector Spaces

A **vector space** is a set of vectors where addition and scalar multiplication are defined and obey consistent rules.

## Abelian Group

Vectors form an abelian group under addition — commutative and associative.

## Scalar Multiplication

Scalars distribute over vector addition: $a(\vec{u} + \vec{v}) = a\vec{u} + a\vec{v}$.

## Example: $\mathbb{R}^n$

The n-dimensional real space is the most common vector space in engineering.

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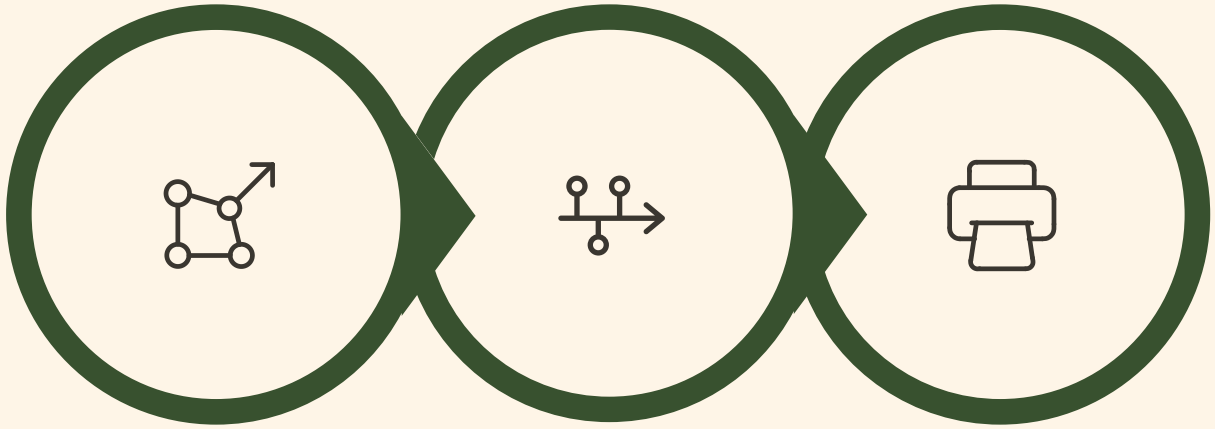
# Linear Transformations

A **linear transformation** maps vectors from one space to another while preserving vector addition and scalar multiplication.



Input Vector

Linear Transformation

Output Vector

These two properties — $L(\vec{u}+\vec{v}) = L(\vec{u}) + L(\vec{v})$ and $L(a\vec{v}) = aL(\vec{v})$ — are what make a transformation *linear*. They are the foundation of all **matrix operations** and geometric manipulations like rotation, scaling, and projection.

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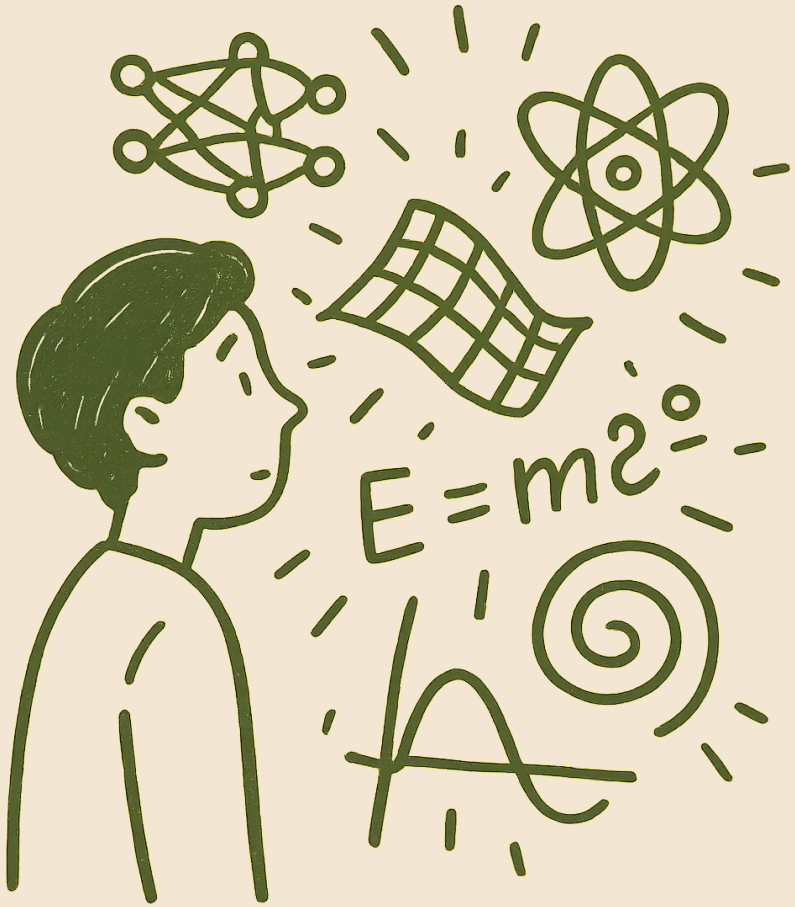


# Your Toolkit for the Future

Mastering these concepts unlocks the door to the most powerful technologies of our era.

### 🤖 AI & ML

Neural networks, gradient descent, and embeddings all live in vector spaces.

### 📊 Data Analysis

PCA, regression, and clustering are linear algebra in action.

### 🎮 Graphics & Vision

Rotations, projections, and 3D rendering rely on transformations and matrices.

**Up Next:** Matrices, determinants, eigenvalues, and eigenvectors — where theory meets real-world computation.

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